# Kerr nonlinearity without modulational instability: ZZ-coupled plasma-wave spin chains in the few-quanta regime

**Anonymous authors**

## Abstract

Spin chains whose one-excitation sector reproduces the dispersion of electromagnetic waves in a plasma can be run on present-day superconducting processors. Ising ($ZZ$) couplings are the natural route from these linear models to nonlinear plasma physics, but it has been unclear which nonlinear phenomenon a given coupling pattern encodes and whether a lattice of ten or so qubits can display it quantitatively. We introduce a normal-ordered next-nearest-neighbour coupling, $K_{j,j+2} = \frac{\epsilon}{2}(-1)^j(\Delta_j + \Delta_{j+2})$, staggered like the plasma-frequency term. It acts as a density-dependent reduction of the local plasma frequency (the lattice analogue of relativistic and ponderomotive self-focusing), leaves the linear physics exactly invariant and vanishes identically at $\epsilon = 0$. A symbolic Schrieffer–Wolff and coherent-state derivation maps the model onto the focusing nonlinear Schrödinger equation (NLSE) with Kerr coefficient $Q = J^2/2\Delta - 2\epsilon\Delta$. The $J^2/2\Delta$ part is a defocusing kinematic interaction generated by the two-level nature of the qubits; it is absent from the naive classical-spin limit and sets a modulational-instability (MI) threshold $\epsilon_c = J^2/4\Delta^2$. Pre-registered full-Hilbert-space emulation on $N = 11$ qubits gives three results. It recovers the linear limit to machine precision; the published noiseless dispersion markers of the reference experiment carry an offset that no ideal circuit reproduces, which we document. It reproduces the Kerr coefficient within $12.7\,\%$ or better at $4$ of $6$ couplings, including the kinematic term alone at $\epsilon = 0$. It fails the pre-registered MI test at all $5$ couplings: no sideband grows, whereas the NLSE predicts e-folding within $5.7$–$42.4\,\hbar/J$. Sector-resolved analysis traces the failure to the few-quanta regime. With $0.6$ wave quanta on average and $59\,\%$ of quanta having no partner, the parametric amplification behind MI cannot develop. Reaching the NLSE regime requires $\gtrsim 10^2$ qubits, beyond classical statevector emulation.

## 1 Introduction

Nonlinear wave phenomena, such as self-modulation, self-focusing, parametric scattering and induced transparency, govern how intense electromagnetic (EM) waves propagate in plasmas, from inertial-fusion targets to pulsar magnetospheres [1, 2, 3, 4]. Their quantum many-body counterparts, relevant to strongly coupled and relativistic quantum plasmas, are hard to simulate classically. This has motivated a programme of quantum algorithms for plasma physics [5, 6, 7, 8, 9], building on quantum simulation of field theories more broadly [10, 11], and of quantum-hardware demonstrations of wave–wave interactions [12, 13]. A particularly hardware-friendly building block is the spin-chain model of plasma waves of Sundar *et al.* [14]. A chain of qubits with an $XY$-type coupling $J$ and a staggered longitudinal field $\Delta_j$ has one-excitation eigenstates with dispersion $\hbar\omega_k = \pm\sqrt{\Delta^2 + J^2 \sin^2 ka}$. This is the lattice version of the EM dispersion relation $\omega^2 = \omega_p^2 + c^2k^2$, with $\Delta \leftrightarrow \hbar\omega_p$, and Trotterised evolution needs only nearest-neighbour gates. That model is, however, linear: its one-excitation sector is a free-particle problem and can be simulated efficiently on a classical computer.

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Adding Ising ($ZZ$) couplings is the obvious way to make the model nonlinear and classically hard, and the original authors explicitly anticipated it [14]. Two questions are open. First, *which* $ZZ$ coupling pattern encodes *which* classical nonlinear plasma equation, with which coefficients? Second, can the nonlinear phenomenon be demonstrated quantitatively on the $\sim 10$-qubit lattices accessible to present hardware and to exact classical emulation? This paper answers both for one carefully chosen coupling, with a positive answer to the first question and a largely negative, mechanistically explained answer to the second.

**Contributions.**

- **A density-dependent plasma-frequency coupling** (Sec. 2). We add $H_{ZZ} = \sum_j K_{j,j+2}\, n_j n_{j+2}$ with $K_{j,j+2} = \frac{\epsilon}{2}(-1)^j(\Delta_j + \Delta_{j+2})$ and $n_j = (1 - Z_j)/2$. Since $H_{ZZ}$ is normal ordered with respect to the spin-wave vacuum, it leaves the vacuum and every one-quantum state untouched, so the linear plasma-wave physics is identical for all $\epsilon$. It reduces the local plasma frequency in proportion to the local density of wave quanta, which is the lattice counterpart of relativistic mass increase and ponderomotive depletion.
- **An explicit map to the focusing NLSE** (Sec. 3). A symbolic derivation (projection onto the upper branch, then spin-coherent mean field, then continuum limit) yields $i\partial_t\psi = -P\partial_x^2\psi + Q|\psi|^2\psi$ with $P = J^2/2\Delta$ and $Q = Q_{\rm kin} + Q_{\rm dyn} = J^2/2\Delta - 2\epsilon\Delta$. The defocusing term $Q_{\rm kin}$ is the hard-core (Dyson kinematic) interaction of the qubit encoding. It is invisible in the naive classical-spin limit, and it sets an instability threshold $\epsilon_c = J^2/4\Delta^2$.
- **Pre-registered full-Hilbert-space emulation** (Secs. 4, 5). On $2^{11}$-dimensional statevectors we find the following.
  - The linear limit is exact at machine precision.
  - The NLSE Kerr coefficient is reproduced within $12.7\,\%$ or better at $4/6$ couplings, including the kinematic term alone.
  - The pre-registered modulational-instability (MI) growth-rate test fails at every one of the $5$ couplings. All points are reported.
- **Mechanism of the breakdown** (Sec. 6). The failure has three ingredients:
  - *few-quanta dilution*: most wave quanta have no partner to interact with, and the one-quantum sector is exactly linear;
  - *finiteness*: a sector with $m$ quanta on six sites cannot sustain parametric amplification;
  - *granularity*: the $ZZ$ phase per pair is periodic, so the mean-field Kerr picture holds only while $|V|t \lesssim 1$.

  We give the scaling condition for the NLSE regime and its consequence for quantum advantage (Sec. 7).

**Related work.** *Modulational instability.* MI of a uniform wave, governed at weak amplitude by the NLSE through the Lighthill criterion [15, 16, 17], is a central plasma phenomenon. It appears for Langmuir waves [18, 19], for cyclotron waves [20], and for relativistically intense EM waves, where the plasma frequency $\omega_p^2/\gamma$ decreases with intensity [21, 22, 3, 1, 2]. Reviews are given in Refs. [23, 24]. Discreteness [25] and nonlocality [26] modify the MI band; both matter on a lattice of six effective sites.

*Spin chains and field theory.* Mapping spin chains to field theories has a long history: Holstein–Primakoff bosons [27], the hard-core boson representation [28], Jordan–Wigner fermions [29], and the correspondence between the anisotropic chain and the massive Thirring and sine-Gordon models [30, 31, 32]. Large-spin limits produce magnetic solitons governed by the NLSE [33]. Our derivation is closest to the Holstein–Primakoff plus classical-limit route. It differs in two ways: it treats the staggered (two-component, Dirac-like) chain near its gap, and it keeps the leading quantum correction, Dyson's kinematic interaction [34], which the classical spin limit drops (Sec. 3). Bosonization is not the appropriate tool here, because we work in the dilute upper-branch regime rather than near half filling. In that regime the relevant continuum theory is the NLSE, the mean-field limit of the Lieb–Liniger gas [35, 36, 37].

*Quantum versus classical nonlinear dynamics.* The finite-quanta physics we encounter (bound magnons [38, 39, 40], collapse and revival of coherent fields [41], the domain of validity of truncated-

Wigner and mean-field descriptions [42, 43]) is familiar from cold atoms, including quench experiments on tilted spin states in tunable Heisenberg chains [44] and discrete nonlinear dynamics in condensate arrays [45]. There MI has been observed in condensates with $10^3$–$10^5$ atoms [46, 47, 48].

*Quantum algorithms for nonlinear equations.* Our setting differs from algorithms that solve classical nonlinear PDEs by Carleman, Koopman or Lax-pair embeddings [49, 50, 51, 52, 53], and from recent symplectic formulations [54]. Here the quantum dynamics of the spin chain *is* the object of interest, and the classical PDE is its emergent mean-field limit.

## 2 A spin-chain model with a density-dependent plasma frequency

**Linear model.** We keep the plasma-wave chain of Ref. [14] unchanged ($\hbar = a = 1$; sites $j = 1, \dots, N$):

$$H_0 = -\frac{J}{4}\sum_{j=1}^{N-1}\left(X_jY_{j+1} - Y_jX_{j+1}\right) + \frac{1}{2}\sum_{j=1}^{N}\Delta_j(-1)^j Z_j. \tag{1}$$

With $Z|0\rangle = |0\rangle$, a wave quantum is a qubit in $|1\rangle$. In terms of the hard-core boson $a_j = |0\rangle\langle 1|_j$ and $n_j = a_j^\dagger a_j = (1 - Z_j)/2$ [28],

$$H_0 = \sum_j \left(\tfrac{iJ}{2}a_j^\dagger a_{j+1} + \text{h.c.}\right) - \sum_j \Delta_j(-1)^j n_j + \text{const.} \tag{2}$$

Odd sites (sublattice $A$) carry on-site energy $+\Delta_j$ and even sites ($B$) carry $-\Delta_j$. In a uniform plasma the two-site Bloch Hamiltonian $\left(\begin{smallmatrix}\Delta & -J\sin k\\ -J\sin k & -\Delta\end{smallmatrix}\right)$ gives the two branches $\pm\sqrt{\Delta^2 + J^2\sin^2 k}$ (Fig. 1, right). Near $k = 0$ the upper, positive-frequency branch lives almost entirely on sublattice $A$: its $B$ amplitude is $\mathcal{O}(Jk/2\Delta)$. The $A$ and $B$ components play the roles of the two components of a discretised Dirac spinor [14], so the physical field lives on a grid of spacing $2a$.

**The $ZZ$ extension.** We commit to a single coupling matrix (Fig. 1):

$$H(\epsilon) = H_0 + H_{ZZ}, \qquad H_{ZZ} = \sum_{j=1}^{N-2} K_{j,j+2}\, n_j n_{j+2} = \frac{1}{4}\sum_j K_{j,j+2}\left(Z_jZ_{j+2} - Z_j - Z_{j+2} + 1\right), \tag{3}$$

$$K_{j,j+2}(\epsilon) = \frac{\epsilon}{2}(-1)^j\left(\Delta_j + \Delta_{j+2}\right), \qquad K_{il} = 0 \ \text{ for } |i-l| \neq 2. \tag{4}$$

The construction rests on four properties.

1. *Physical content.* Combining Eq. (3) with the mass term of Eq. (2) gives $-\sum_j (-1)^j \Delta_j^{\text{eff}} n_j$, where in a uniform plasma $\Delta_j^{\text{eff}} = \Delta[1 - \frac{\epsilon}{2}(n_{j-2} + n_{j+2})]$. The local plasma frequency is therefore lowered in proportion to the local density of wave quanta. This is the lattice analogue of the relativistic ($\omega_p^2 \to \omega_p^2/\gamma$) and ponderomotive (electron-depletion) nonlinearities of intense EM waves, both of which reduce $\omega_p$ where the intensity is high [21, 3, 1, 2]. Like these nonlinearities, $H_{ZZ}$ is proportional to the local plasma frequency and vanishes in vacuum ($\Delta_j = 0$).
2. *Locality.* The couplings join next-nearest neighbours of the chain, i.e. nearest neighbours *within* a sublattice. In the continuum limit this is a local self-interaction of each spinor component. Nearest-neighbour $ZZ$ would instead couple $A$ to $B$, a Thirring-type current–current interaction [30, 31]. Its Kerr coefficient for an upper-branch wave scales with the $B$ weight, $\propto (Jk/2\Delta)^2$, and vanishes at the plasma cutoff where the self-modulation physics resides. This is why we rejected it (Appendix B).
3. *Sign structure.* The coupling is attractive on the $+\Delta$ sublattice and repulsive on the $-\Delta$ sublattice. The positive- and negative-frequency branches of the real wave field therefore acquire the same focusing nonlinearity, as complex conjugation requires.
4. *Exact linear limit.* $\epsilon$ is an overall prefactor, so $K(0) \equiv 0$ and $H(0) = H_0$ identically. Because the $ZZ$ term is normal ordered, $H_{ZZ}$ annihilates the vacuum and every one-quantum state for *all* $\epsilon$. The linear plasma-wave physics (dispersion, reflection, propagation)

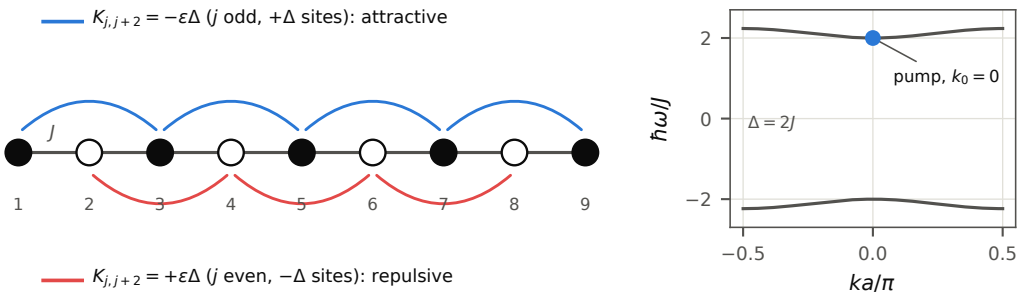



**Figure 1: The ZZ-extended plasma-wave chain.** Left: filled (open) circles are the $+\Delta$ ($-\Delta$) sublattice. Grey bonds are the $XY$ coupling $J$. Arcs are the committed next-nearest-neighbour couplings of Eq. (4) in a uniform plasma: attractive on the $+\Delta$ sublattice, repulsive on the $-\Delta$ sublattice. Right: the two plasma-wave branches $\pm\sqrt{\Delta^2 + J^2 \sin^2 ka}$ for $\Delta = 2J$. The modulational-instability experiments pump the cutoff mode $k_0 = 0$ of the upper branch.

is untouched, and every effect of $\epsilon$ resides in sectors with $\geq 2$ quanta. The single-$Z$ terms in Eq. (3) cancel the Hartree shift that a bare $ZZ$ coupling would impose on single quanta through the polarised background.

On hardware each $n_j n_{j+2}$ term is a controlled phase, the $\phi$ parameter of an FSIM gate [55]. A Trotter step therefore needs $N-1$ $XY$ gates plus $N-2$ controlled phases between next-nearest neighbours, which are adjacent on a zig-zag (triangular-ladder) embedding.

## 3 From the spin chain to the nonlinear Schrödinger equation

The derivation is carried out symbolically in `derive_nlse.py`. Every intermediate expression is written to `logs/derivation/derivation_log.md` (items A–G, cited below), and the resulting coefficients were frozen in the method plan before any emulation with $\epsilon \neq 0$ (Sec. 4). We take a uniform plasma, $\Delta_j = \Delta$.

**Projection onto the upper branch.** For $J \ll 2\Delta$ we eliminate the $B$ sublattice by a second-order Schrieffer–Wolff transformation [56, 57]. A quantum on $A$ site $j$ virtually hops to an empty $B$ neighbour, with energy denominator $2\Delta$, and either returns or continues to $j \pm 2$. The effective Hamiltonian on the $A$ sites $s$, spaced by $2a$, is

$$H_{\text{eff}} = \sum_s (\Delta + t\, z_s)\, n_s - t \sum_s \left(a^\dagger_{s+1} a_s + \text{h.c.}\right) + V \sum_s n_s n_{s+1}, \qquad t = \frac{J^2}{8\Delta}, \quad V = K_{\text{odd}} = -\epsilon\Delta, \tag{5}$$

where $z_s \in \{1, 2\}$ counts the $B$ neighbours. The single-quantum dispersion $\Delta + 2t - 2t\cos 2K$ reproduces the exact upper branch $\sqrt{\Delta^2 + J^2 \sin^2 K}$ to $\mathcal{O}(J^2/\Delta)$ (checked symbolically, item D4). The hard-core constraint enters through the hopping: $a^\dagger_{s+1} a_s$ vanishes when $s+1$ is occupied. It does not enter the on-site term, because the virtual return trip $A \to B \to A$ is never blocked.

**Coherent-state (Hartree) mean field.** We evaluate Eq. (5) in spin-coherent product states, $\psi_s = \langle a_s \rangle = \frac{1}{2} \sin\theta_s e^{i\varphi_s}$ and $n_s = \sin^2(\theta_s/2)$, so that $|\psi_s|^2 = n_s(1 - n_s)$. For a uniform wave the energy per $A$ site is $E(n) = (\Delta + 2t)n - 2t\, n(1-n) + V n^2$. The wave frequency $\omega = dE/dn$ is therefore

$$\omega(n) = \Delta + \big(\underbrace{4t}_{Q_{\text{kin}}} + \underbrace{2V}_{Q_{\text{dyn}}}\big)\, n. \tag{6}$$

The *kinematic* term $Q_{\text{kin}} = 4t = J^2/2\Delta$ is the lattice form of Dyson's kinematical spin-wave interaction [34]. A quantum can only hop onto an empty site, which reduces the kinetic energy by the factor $(1-n)$ and acts as a defocusing (repulsive) Kerr term. It is a property of $H_0$ alone. Remarkably, it is *absent* from the naive classical-spin limit of the full chain (item G). For a uniform $A$-sublattice wave, the $B$ spins feel no transverse field because the two $A$ neighbours interfere destructively, so the

**Table 1: Term-by-term correspondence** between $H = H_0 + H_{ZZ}$ and the NLSE (8), together with the plasma interpretation (weakly relativistic, circularly polarised EM wave near cutoff, $\omega^2 = \omega_p^2/\gamma + c^2k^2$, which gives $i\partial_t a = -(c^2/2\omega_p)\partial_x^2 a - (\omega_p/4)|a|^2 a$).

| Hamiltonian term | NLSE term | plasma counterpart |
|---|---|---|
| $\frac{1}{2}\Delta(-1)^j Z_j$ (staggered mass) | carrier frequency $\Delta$ | cutoff $\hbar\omega_p$ |
| $-\frac{J}{4}(XY - YX)$ via virtual $A \to B \to A$ hops | dispersion $-P\partial_x^2$, $P = J^2/2\Delta$ | $c^2/2\omega_p$, i.e. $Ja/\hbar \leftrightarrow c$ |
| hard-core constraint of $H_0$ | $Q_{\rm kin}\lvert\psi\rvert^2\psi$, $Q_{\rm kin} = J^2/2\Delta > 0$ | none: saturation of the two-level encoding (defocusing) |
| $\frac{1}{4}K_{j,j+2}Z_jZ_{j+2}$ | $Q_{\rm dyn}\lvert\psi\rvert^2\psi$, $Q_{\rm dyn} = 2K_{\rm odd} = -2\epsilon\Delta$ | $-\frac{\omega_p}{4}\lvert a\rvert^2 a$, i.e. $\epsilon n \leftrightarrow \lvert a\rvert^2/8$ |
| $-\frac{1}{4}K_{j,j+2}(Z_j + Z_{j+2})$ | cancels the one-quantum shift $-\frac{1}{2}K$ per bond that bare $ZZ$ would add | none (normal ordering) |
| $\cos 2K$ form factor of $K_{j,j+2}$ | nonlocal (lattice-scale) Kerr kernel | finite-range nonlinear response [26] |

classical spins precess at $\Delta + 2Vn$. The quantum virtual fluctuations captured by the projection are what generate $Q_{\rm kin}$. Section 5 confirms this term in the emulation.

**Linear stability and the NLSE.** Linearising the mean-field dynamics about the uniform wave, with a perturbation $\propto \cos(2Ks)$, gives the exact-amplitude Bogoliubov growth rate (items E4–E5)

$$\Gamma^2 = -\varepsilon_K(1-2n)\big[\varepsilon_K(1-2n)+2G_K\big], \qquad \varepsilon_K = 4t\sin^2 K, \quad G_K = \frac{(4t + 2V\cos 2K)\,n(1-n)}{1-2n}. \tag{7}$$

We verified Eq. (7) against direct integration of the classical-spin equations of motion of $H_{\rm eff}$ on a 48-site ring. The maximum relative deviation is $4.9 \times 10^{-5}$ over 17 unstable cases (`verify_linearisation.py`). To first order in the amplitude and in $K^2$, Eq. (7) is the modulational-instability rate of the focusing NLSE

$$i\partial_t\psi = -P\,\partial_x^2\psi + Q\,|\psi|^2\psi, \qquad P = \frac{J^2}{2\Delta}, \qquad Q(\epsilon) = \frac{J^2}{2\Delta} - 2\epsilon\Delta, \tag{8}$$

written in the frame rotating at $\Delta$ with $|\psi|^2$ the density of wave quanta (items F1–F5). The Lighthill criterion $PQ < 0$ [15] gives instability exactly when

$$\epsilon > \epsilon_c = \frac{J^2}{4\Delta^2}, \tag{9}$$

and a perturbation of wavenumber $K$ on a background of density $n_0$ grows at

$$\Gamma(K) = \sqrt{E_K\,(2|Q|n_0 - E_K)}, \qquad E_K = PK^2, \qquad \Gamma_{\max} = |Q|n_0 \ \text{ at } \ E_K = |Q|n_0. \tag{10}$$

On the emulation grid (the $A$ sites of an open chain with odd $N$, spacing $2a$, reflecting ends) the sideband modes are $\cos(K_p j)$ with $K_p = p\pi/(N+1)$. The central-difference discretisation of Eq. (8) replaces $E_K$ by $E_p = P\sin^2 K_p$. This discretised-NLSE rate is our pre-registered reference.

**Where the map can fail.** The derivation identifies five controlled approximations and hence five predicted failure modes, which we test in Secs. 5–6.

(i) *Mean-field factorisation*, which requires many quanta per interaction volume.

(ii) *Lattice-scale wavenumbers*, where the form factor $\cos 2K$ of the next-nearest-neighbour kernel departs from 1. In Eq. (7) it can stabilise modes the local NLSE calls unstable.

(iii) *Finite amplitude*, through the factors $(1-2n)$ and $n$ versus $|\psi|^2$.

(iv) *Higher orders in $J/2\Delta$ and $\epsilon n$* in the Schrieffer–Wolff expansion.

(v) *Trotter error* in the emulated circuit.

## 4 Emulation and pre-registered protocol

**Full-Hilbert-space emulation.** States are stored as all $2^N$ complex amplitudes and nothing is ever truncated. Each first-order Trotter step [58, 59] is $U(\delta) = D\,U_{\mathrm{even}}U_{\mathrm{odd}}$, with layers in the same order as the reference experiment [14]:

- $U_{\mathrm{odd}}$ and $U_{\mathrm{even}}$ apply $\exp(-i\delta h_b)$ exactly on the bonds $(1,2),(3,4),\ldots$ and $(2,3),(4,5),\ldots$ respectively, where $h_b$ is the $XY$ term of bond $b$ (a real rotation inside $\mathrm{span}\{|01\rangle,|10\rangle\}$);
- $D = \exp[-i\delta(H_Z + H_{ZZ})]$ is diagonal and applied exactly.

Because $H_{ZZ}$ enters only through $D$, the $\epsilon = 0$ circuit is bit-for-bit the $H_0$ circuit. Exact evolution by sector-wise diagonalisation ($H$ conserves $\sum_j n_j$) provides an independent check of every Trotter run. The emulator is a NumPy implementation written for this study (`spinchain.py`).

**Pre-registration.** The study followed a written plan frozen in three stages. Each stage was hashed (SHA-256) into `logs/method_plan_commits.log` before the runs that depend on it.

- *Stage 0* (2026-10-05 03:32:17 UTC), before any emulation with $\epsilon \neq 0$: the coupling matrix (4), the target phenomenon (modulational instability of a cutoff wave), the PDE family (focusing NLSE), the observable class (growth rate of the most unstable sideband) and the tolerance ($\leq 15\,\%$ relative error).
- *Stage 0b* (2026-10-05 03:36:18 UTC): the derived coefficients $P$, $Q$, $\epsilon_c$ and the exact reference formula, i.e. the discretised NLSE of Sec. 3 with $|\psi_0|^2 \equiv n_0$.
- *Stage 1* (2026-10-05 03:47:06 UTC), after two disclosed pilot studies ($36 + 6$ configurations, retained in `logs/pilot/`): the protocol, the $\epsilon$ grids, and one *additional* observable.

The pilots had already shown that no sideband grows in the quantum emulation. Stage 1 therefore kept the MI test exactly as registered and added, before any of its validation runs, a secondary test of the NLSE nonlinear coefficient itself. Neither the coupling matrix, nor the phenomenon, nor the PDE reference, nor the tolerance was changed. Every declared coupling is reported.

**Common settings.** $N = 11$ (odd, so that the uniform $A$-sublattice wave is an exact $k_0 = 0$ eigenmode of $H_0$; $L_A = 6$ sites), $J = 1$, $\Delta = 2J$. This gives $t = J/16$, $P = 0.25$ and $\epsilon_c = 0.0625$. We use Trotter step $\delta = 0.1$. The dispersion and the kinematic term arise from virtual processes at frequency $2\Delta = 4J$, which the step $\delta = 0.8$ used for the linear experiments of Ref. [14] would alias ($2\Delta\delta = 3.2$ rad). We therefore also report $\delta = 0.8$ and exact evolution for every point.

**Modulational instability (primary).**

- *Pump*: every $A$ qubit in the spin-coherent state of density $n_0 = 0.1$ (a uniform upper-branch wave at cutoff); $B$ qubits in $|0\rangle$.
- *Seed*: the $A$-site field is multiplied by $1 \pm \eta\cos(K_p j)$ with $\eta = 0.05$, for each sideband $p = 1,\ldots,5$.
- *Observable*: the linear response of the sideband density, $R_p(t) = [M_p^+(t) - M_p^-(t)]/[M_p^+(0) - M_p^-(0)]$, where $M_p = \sum_{j\in A}\cos(K_p j)\langle n_j\rangle / \sum_{j\in A}\cos^2(K_p j)$. The NLSE predicts $R_p = \cosh(\Gamma_p t)$.
- *Extraction*: a least-squares fit of $R_p = \cosh(\sqrt{g}\,t)$ with $g \in \mathbb{R}$ (a cosine for $g < 0$) over $0 \leq t \leq \mathrm{arccosh}(4)/\Gamma_p$, the time of four-fold NLSE growth. Windows range from $11.8$ to $87.4\,\hbar/J$. The growth rate is $\Gamma_{\mathrm{emu}} = \sqrt{\max(g,0)}$.
- *Grid*: $\epsilon \in \{0.125, 0.1875, 0.25, 0.375, 0.5\}$, i.e. 2–8 $\epsilon_c$. The comparison is made for the most unstable sideband $p^*$.

The same seeds are evolved with the discretised NLSE (RK4, `pde_reference.py`) and with $H_0$.

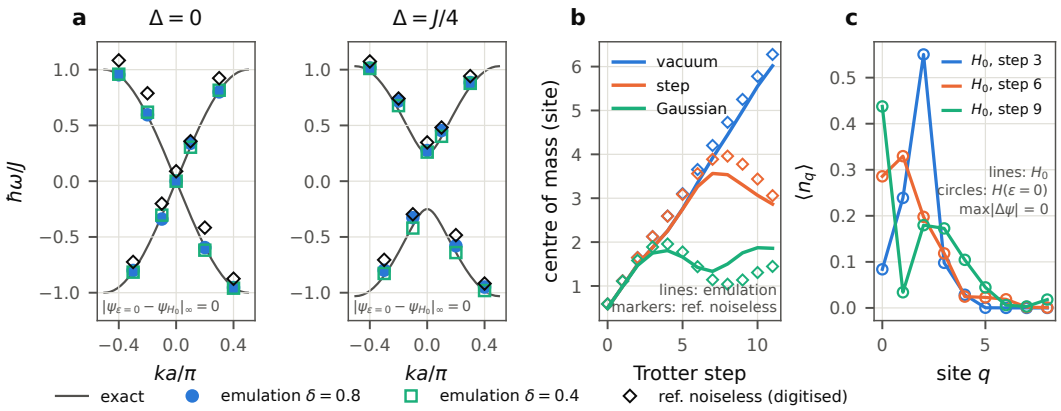



**Figure 2: Linear limit at $\epsilon = 0$ ($N = 9$). a** Ramsey dispersion. Grey curves: exact branches. Filled (open) symbols: our emulation with $\delta = 0.8$ (0.4). Diamonds: the reference paper's noiseless markers, digitised. The $\epsilon = 0$ extended circuit coincides with $H_0$ bit for bit (inset text). **b** Wave-packet centre of mass in vacuum, at a sharp boundary and in a Gaussian plasma. Lines: emulation; diamonds: reference noiseless traces. **c** Site densities in the Gaussian plasma after 3, 6 and 9 steps. Lines: $H_0$; circles: $H(\epsilon = 0)$. The maximum statevector difference is exactly zero.

**Kerr coefficient (secondary).** The unseeded pump is evolved for $n_0 \in \{0.02, 0.05, 0.10\}$ and for a linear reference $n_0 = 10^{-4}$. The frequency $\omega$ is minus the slope of the unwrapped phase of $\langle a_5 \rangle + \langle a_7 \rangle$ (the two central $A$ sites), fitted over $0 \le t \le 2\,\hbar/J$. $Q_{\mathrm{emu}}$ is the least-squares slope through the origin of $\delta\omega(n_0) = \omega(n_0) - \omega(10^{-4})$ versus $n_0$. The grid is $\epsilon \in \{0, 0.125, 0.1875, 0.25, 0.375, 0.5\}$; the point $\epsilon = 0$ isolates $Q_{\mathrm{kin}}$ and doubles as the $H_0$ contrast. The plan stated in advance that agreement could only be expected while the per-pair $ZZ$ phase $|V|t$ stays below about one.

**Linear limit.** We reproduce both linear experiments of Ref. [14] at exactly $\epsilon = 0$ with $N = 9$.

- The Ramsey dispersion protocol: qubit 1 in $(|0\rangle + |1\rangle)/\sqrt{2}$, 12 steps, with $\Delta = 0$ and $J/4$.
- Wave-packet propagation in vacuum, at a sharp plasma boundary ($\Delta = J$ on sites 6–9), and through a Gaussian plasma.

The reference curves were digitised from the vector graphics of the published PDF. The axis-fit residual is $\le 4.3 \times 10^{-5}$ in units of $\hbar\omega/J$, and no value was entered by hand (`paper_reference.py`).

## 5 Results

### 5.1 Linear limit

Setting $\epsilon = 0$ in Eq. (4) gives $K \equiv 0$ symbolically (derivation item A3), and the emulated circuits are then identical to the $H_0$ circuits element by element. The largest statevector difference over all dispersion and propagation runs is exactly 0, and the Hamiltonians differ by 0. Because $H_{ZZ}$ is normal ordered, the one-quantum observables coincide with $H_0$ at *every* finite $\epsilon$ of our sweeps: the largest difference in dispersion frequencies, centre-of-mass traces or states is 0.

**Dispersion.** Exact evolution reproduces all nine eigenfrequencies to machine precision (largest error 0.0 %; Table 4 in Appendix C). With Trotter step $\delta = 0.8$ the extracted frequencies carry Trotter error: at most 9.9 % for $\Delta = 0$ and 17.7 % for $\Delta = J/4$, reduced to 5.2 % and 6.1 % at $\delta = 0.4$.

The reference paper's noiseless markers lie systematically above our $\delta = 0.8$ values, by $0.114\,J$ ($\Delta = 0$) and $0.054\,J$ ($\Delta = J/4$) on average, and above the exact branches as well (Table 4). Only $0/9$ and $2/9$ of them are within 5 % of our emulation. No ideal circuit can produce this offset at $\Delta = 0$. In the one-quantum sector each layer of $U(\delta)$ is a real orthogonal matrix, so the Trotterised spectrum is symmetric about zero and contains an exact zero mode for odd $N$. The reference marker of that mode sits at $0.089\,J$. The reference wave packet likewise starts at a centre of mass of 0.59 sites, whereas the stated two-site state has 0.5. We conclude that the reference "noiseless" markers

**Table 2: Kerr coefficient from self-phase modulation** (secondary, pre-registered; $\Delta = 2J$, $N = 11$, fit window $T = 2\,\hbar/J$). $|V|T$ is the $ZZ$ phase accumulated per neighbouring pair. "$H_0$" is the $\epsilon = 0$ value (contrast). "cl. spin" is the naive classical-spin limit of the full chain.

| $\epsilon$ | $\epsilon/\epsilon_c$ | $\lvert V\rvert T$ | $Q_{\mathrm{NLSE}}$ | $Q_{\mathrm{emu}}$ ($\delta$=0.1) | exact | $\delta$=0.8 | $H_0$ | cl. spin | err. (%) | $\leq 15\%$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0.00 | 0.250 | 0.238 | 0.242 | −0.018 | 0.238 | 0.000 | 4.7 | pass |
| 0.125 | 2 | 0.50 | −0.250 | −0.224 | −0.221 | −0.433 | 0.238 | −0.500 | 10.5 | pass |
| 0.1875 | 3 | 0.75 | −0.500 | −0.447 | −0.445 | −0.626 | 0.238 | −0.750 | 10.5 | pass |
| 0.25 | 4 | 1.00 | −0.750 | −0.655 | −0.653 | −0.804 | 0.238 | −1.000 | 12.7 | pass |
| 0.375 | 6 | 1.50 | −1.250 | −0.987 | −0.987 | −1.093 | 0.238 | −1.502 | 21.1 | fail |
| 0.5 | 8 | 2.00 | −1.750 | −1.162 | −1.164 | −1.263 | 0.238 | −2.003 | 33.6 | fail |

include features of the compiled hardware circuit that are not specified in the publication. Neither the stated step ($\delta = 0.8$) nor the step suggested by visual inspection of the reference phase plots ($\approx 0.4$) reproduces them within $5\,\%$. We therefore report the comparison and do not tune toward it.

**Propagation.** The centre-of-mass traces agree qualitatively with the reference in all three geometries. The first reflection occurs at step 11, 7 and 4 in our emulation versus 11, 8 and 4 in the reference (vacuum, step, Gaussian). The maximum penetration is 6.02, 3.56 and 1.81 sites versus 6.28, 3.96 and 1.96. The rms deviations are 0.26, 0.30 and 0.32 sites. Reversing the order of the two $XY$ layers changes them to 0.28, 0.18 and 0.15, which is comparable to the initial offset.

### 5.2 The nonlinear coefficient: self-phase modulation of the pump

Table 2 and Fig. 3b test the coefficient $Q(\epsilon)$ of Eq. (8) directly. Four observations follow.

1. *The kinematic term is real.* With $H_0$ alone the pump frequency rises with density, $Q_{\mathrm{emu}} = 0.238$, against the derived $Q_{\mathrm{kin}} = 0.25$ ($4.7\,\%$). The classical-spin limit of the same chain predicts 0.000. The quantum emulation therefore follows the Schrieffer–Wolff-corrected NLSE, not the naive classical field theory. This is a non-trivial, quantitatively confirmed prediction of the mapping.

2. *The $ZZ$ term reverses the sign of the nonlinearity.* Above $\epsilon_c$ the shift becomes negative (focusing) and follows $Q_{\mathrm{NLSE}} = J^2/2\Delta - 2\epsilon\Delta$. Agreement is within $10.5\,\%$, $10.5\,\%$ and $12.7\,\%$ at $\epsilon = 0.125$, 0.1875 and 0.25. In total 4 of 6 pre-registered points pass.

3. *The two strongest couplings fail*, by $21.1\,\%$ and $33.6\,\%$. There $|V|T = 1.50$ and 2.00, as anticipated in the plan (Sec. 6). The deficit is not an amplitude effect: $\delta\omega/n_0$ is constant to within a few per cent for $0.005 \leq n_0 \leq 0.2$ (from $-0.659$ to $-0.638$ at $\epsilon = 0.25$).

4. *Trotterisation matters.* $\delta = 0.8$ gives $-0.018$ at $\epsilon = 0$ (the wrong sign), whereas $\delta = 0.1$ lies within $2.5\,\%$ of exact evolution (Fig. 6a).

### 5.3 Modulational instability

The pre-registered instability test fails at all 5 couplings (Table 3, Fig. 3a). The discretised NLSE, solved numerically from the same seeds, grows exactly as predicted (largest deviation from the analytic rate $1.3\,\%$). It reaches $R_{p^*} \approx 3.98$ at the end of the window at $\epsilon = 0.25$.

The full-statevector emulation of $H_0 + H_{ZZ}$ instead gives $R_{p^*} = 0.15$ there, and its best fit is oscillatory ($g < 0$) at every coupling, so $\Gamma_{\mathrm{emu}} = 0$ and the relative error is $100\,\%$. Exact evolution and $\delta = 0.8$ agree. Every one of the 25 (coupling, sideband) pairs, not only $p^*$, is listed in Table 5.

The $H_0$ contrast (Fig. 4a) shows that the $ZZ$ term does act in the predicted direction, only far too weakly. For the most unstable sideband at $\epsilon = 0.25$ the fitted $g = -2.9 \times 10^{-3}$ is less negative than its $H_0$ value $-4.9 \times 10^{-3}$: the sideband softens. The NLSE predicts $g = +\Gamma^2$, however, and $g$ never changes sign. Self-phase modulation of the same pump, in contrast, distinguishes $H$ from $H_0$ cleanly (Fig. 4b): the $ZZ$ term turns the defocusing kinematic phase into a focusing one of the predicted slope.

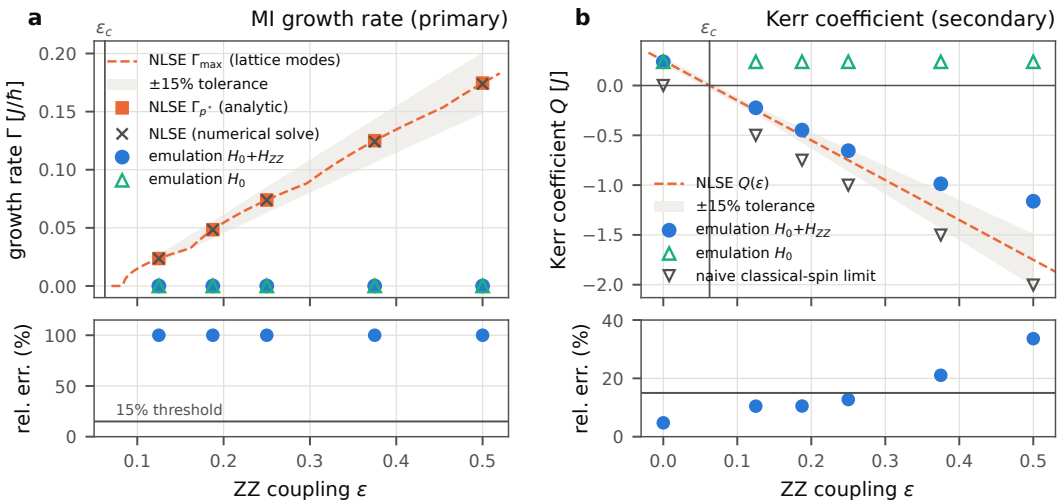



**Figure 3: Emulation versus the NLSE across the pre-registered $\epsilon$ sweeps. a** Modulational instability. Squares: NLSE growth rate of the most unstable lattice sideband; crosses: numerical solution of the same discretised NLSE; band: $\pm 15\,\%$; dashed: NLSE envelope over the five lattice sidebands. Filled circles: full-statevector emulation of $H_0 + H_{ZZ}$; triangles: $H_0$. No emulated sideband grows. The lower panel gives the relative error per point against the $15\,\%$ threshold. **b** Kerr coefficient from self-phase modulation. The emulation (circles) follows $Q_{\mathrm{NLSE}}(\epsilon)$ up to $\epsilon = 0.25$, including the sign change at $\epsilon_c$. The $H_0$ contrast (triangles) stays at the kinematic value. The naive classical-spin limit (open down-triangles) misses the kinematic term.

**Table 3: Main results** for every pre-registered coupling. Primary observable: growth rate $\Gamma$ (units of $J/\hbar$) of the most unstable sideband $p^*$, with emulation ($\delta = 0.1$; exact evolution and $\delta = 0.8$ give the same $\Gamma_{\mathrm{emu}}$) against the discretised NLSE. Linear-limit columns: Trotter error of the $\epsilon = 0$ dispersion against exact eigenfrequencies ($\Delta = 0\,/\,J/4$); rms centre-of-mass deviation from the reference traces (vacuum/step/Gaussian, in sites); the number of reference noiseless dispersion markers within $5\,\%$; and, at $\epsilon > 0$, the largest difference of one-quantum observables from $H_0$. Notes: Kerr coefficient $Q_{\mathrm{emu}}$ vs $Q_{\mathrm{NLSE}}$ (secondary).

| $\epsilon$ | $\Gamma_{\mathrm{emu}}$ | $\Gamma_{\mathrm{NLSE}}$ | rel. err. | linear-limit observable | linear-limit check | notes (Kerr, secondary) |
|---|---|---|---|---|---|---|
| 0 | n/a | n/a | n/a | exact evol.: 0.0 %; Trotter $\delta$=0.8: 9.9 % / 17.7 %; CoM rms 0.26/0.30/0.32 | $H(0) \equiv H_0$ bitwise; ref. markers: 0/9, 2/9 | $Q$: 0.238 vs 0.250 (4.7 %, pass) |
| 0.125 | 0.000 ($p^*$=1) | 0.0236 | 100 % (fail) | 1-quantum $\lvert\Delta\rvert$ vs $H_0$: 0 | identical to $H_0$ | $Q$: $-0.224$ vs $-0.250$ (10.5 %, pass) |
| 0.1875 | 0.000 ($p^*$=2) | 0.0484 | 100 % (fail) | 1-quantum $\lvert\Delta\rvert$ vs $H_0$: 0 | identical to $H_0$ | $Q$: $-0.447$ vs $-0.500$ (10.5 %, pass) |
| 0.25 | 0.000 ($p^*$=2) | 0.0740 | 100 % (fail) | 1-quantum $\lvert\Delta\rvert$ vs $H_0$: 0 | identical to $H_0$ | $Q$: $-0.655$ vs $-0.750$ (12.7 %, pass) |
| 0.375 | 0.000 ($p^*$=3) | 0.1250 | 100 % (fail) | 1-quantum $\lvert\Delta\rvert$ vs $H_0$: 0 | identical to $H_0$ | $Q$: $-0.987$ vs $-1.250$ (21.1 %, fail) |
| 0.5 | 0.000 ($p^*$=4) | 0.1746 | 100 % (fail) | 1-quantum $\lvert\Delta\rvert$ vs $H_0$: 0 | identical to $H_0$ | $Q$: $-1.162$ vs $-1.750$ (33.6 %, fail) |

## 6 Why the instability does not appear: mechanism and breakdown of the map

The analyses in this section were designed after the validation runs. They are diagnostic, not pre-registered, and they use the same logged trajectories plus dedicated runs (`sector_analysis.py`, `failure_analysis.py`, `verify_linearisation.py`).

**Few-quanta dilution.** The pump is a product of spin-coherent states with $n_0 = 0.1$ on $L_A = 6$ sites. It contains on average $N_q = n_0 L_A = 0.6$ wave quanta, distributed binomially: $P(m{=}1) = 35.4\,\%$ and $P(m{\geq}2) = 11.4\,\%$. Since $H$ conserves $m$, every density observable splits exactly into sector contributions. A given quantum has no partner with probability $(1 - n_0)^{L_A - 1}$, which is $59\,\%$. In the seeded experiment $66\,\%$ of the initial sideband modulation sits in the $m = 1$ sector, which evolves under $H_0$ alone: its response differs from the $H_0$ run by exactly 0 (Fig. 5a). The classical NLSE spreads the nonlinearity over the mean density and applies it to every quantum. The quantum dynamics applies it only to the minority of quanta that actually have partners. This is the same

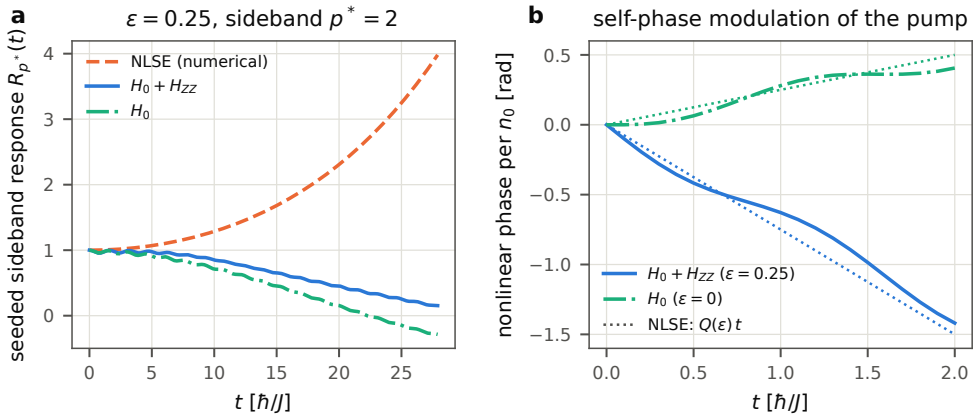



**Figure 4: Contrast between $H = H_0 + H_{ZZ}$ and $H_0$ at $\epsilon = 0.25$.** **a** Seeded response of the most unstable sideband. The discretised NLSE grows as $\cosh \Gamma t$; neither $H$ nor $H_0$ grows. **b** Nonlinear phase of the pump per unit density, $[\phi(n_0{=}0.1) - \phi(0.02)]/0.08$. $H_0$ accumulates the defocusing kinematic phase $Q_{\rm kin}t$, whereas $H$ accumulates the focusing phase $Q(\epsilon)t$. Dotted lines: NLSE.

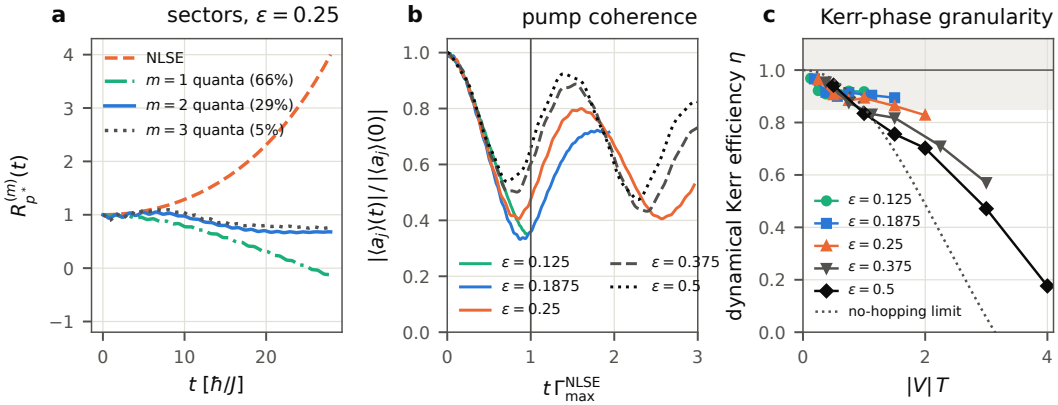



**Figure 5: Mechanism of the breakdown** ($\Delta = 2J$, $n_0 = 0.1$, $N = 11$). **a** Seeded response of the most unstable sideband at $\epsilon = 0.25$, resolved by the number $m$ of wave quanta (percentages: share of the initial modulation). The $m = 1$ sector is exactly linear and $\epsilon$-independent. No sector grows like the NLSE. **b** Coherence of the pump field on the central $A$ sites against time in units of the NLSE e-folding time. By $t = 1/\Gamma_{\max}$ it has fallen to 0.35–0.66 of its initial value. **c** Efficiency $\eta = [Q_{\rm emu}(\epsilon, T) - Q_{\rm emu}(0, T)]/2V$ of the dynamical Kerr term, for fit windows $T = 0.5$–$4\,\hbar/J$, plotted against the $ZZ$ phase per neighbouring pair $|V|T$. All couplings approximately collapse onto one curve that falls below 0.85 for $|V|T \gtrsim 1.5$. Dotted: the no-hopping (pure-Ising) limit, $\mathrm{slope}[\sin(xs)]/x$.

factorisation that makes the *first-order* frequency shift exact (Sec. 5.2) and the *iterated* nonlinear dynamics wrong.

**Finite sectors cannot amplify.** Even the multi-quantum sectors do not grow: over the whole sweep the largest sector response is $\max R^{(m\geq 2)} = 1.21$, against the NLSE's four-fold growth. A sector with $m$ quanta on six sites is a small Hermitian problem with a real spectrum. Seeded sidebands beat against the pump at the frequencies of its few eigenstates (bound magnon clusters and their scattering continuum, cf. [38, 39, 40]) instead of being parametrically amplified. Exponential MI growth is an emergent large-$m$ property. It requires a pump reservoir that loses a negligible fraction of its quanta per created sideband pair (the regime of number-conserving Bogoliubov theory [60]), which a pump holding $\sim 1$ quantum cannot be.

**The pump dephases within one e-fold.** The coherent field of the pump undergoes a collapse-type dephasing, the lattice analogue of Ref. [41]. By the NLSE e-folding time $1/\Gamma_{\max}$, its amplitude on

the central sites has fallen to $0.35$–$0.66$ of its initial value at every coupling (Fig. 5b). The phase reference that MI needs to lock the sideband pair to the pump is lost before amplification can act.

**Granularity of the quantum Kerr phase.** For a quantum with a neighbouring partner of occupation probability $n_0$, the $ZZ$ phase is periodic in time: $\arg(1 - n_0 + n_0 e^{-iVt}) \approx -n_0 \sin Vt$, which reduces to the mean-field $-n_0 Vt$ only for $|V|t \ll 1$. The efficiency $\eta$ of the dynamical Kerr term, extracted from fit windows $T$ between $0.5$ and $4\,\hbar/J$, collapses approximately onto a single function of $|V|T$ (Fig. 5c). It falls from $0.97$ at $|V|T = 0.125$ to $0.83$ at $|V|T = 1.0$ and to $0.18$ at $|V|T = 4.0$. It stays above the pure-Ising curve because hopping motionally narrows the phase. This explains the two failed points of Table 2 and implies a general obstruction for short-range couplings.

The NLSE growth rate is bounded by $\Gamma \leq \Gamma_{\max} = |Q|n_0 < 2|V|n_0$. Over one e-fold each pair therefore accumulates a $ZZ$ phase $|V|/\Gamma > 1/(2n_0)$, which is large precisely in the small-amplitude regime $n_0 \ll 1$ where the NLSE is derived. Hard-core qubits with short-range $ZZ$ couplings can approach the mean-field regime only collectively, through many quanta per healing length. In the language of the dilute Bose gas this means a small Lieb–Liniger parameter [35]; with $\lesssim 3$ quanta on six sites this is out of reach.

**Lattice nonlocality and finite amplitude (classical failure modes).** Separately from quantum effects, the local NLSE is not an accurate classical description of this lattice at its most unstable sidebands. Those sidebands lie at lattice-scale wavenumbers ($K_{p^*} = \pi/12$ to $\pi/3$), where the form factor $\cos 2K$ of the next-nearest-neighbour kernel and the amplitude factor $1 - 2n_0$ in Eq. (7) matter. At $p^*$ the exact-amplitude classical rate is $0.0193$, $0.0000$, $0.0177$, $0.0000$ and $0.0000$, against the NLSE values $0.0236$–$0.1746$. The classical spin limit moves the instability to the longest-wavelength sidebands ($p = 1$–$2$), with rates up to $0.0771$. Even there the quantum emulation shows no growth: its fitted $g$ is negative, $-3.5 \times 10^{-4}$ and $-3.8 \times 10^{-3}$ at $\epsilon = 0.25$ and $0.5$. Thus lattice nonlocality and finite amplitude alone would already have pushed the pre-registered comparison beyond the $15\,\%$ tolerance at 5 of the 5 couplings (deviations of 18–100 $\%$ at $p^*$). The quantum suppression, however, is stronger still and acts at every mode.

**Trotter error.** The kinematic nonlinearity and the dispersion both originate in virtual inter-branch processes at frequency $2\Delta$. A Trotter step must resolve these: at $\epsilon = 0$ the error of the extracted Kerr coefficient is $107.5\,\%$, $30.5\,\%$, $8.9\,\%$, $2.5\,\%$ and $0.4\,\%$ for $\delta = 0.8$, $0.4$, $0.2$, $0.1$ and $0.05$ (Fig. 6a). A hardware demonstration of plasma-wave nonlinearity near cutoff therefore needs $2\Delta\delta \ll 1$, i.e. deeper circuits than the linear experiments of Ref. [14].

**When should the NLSE regime appear?** The NLSE limit requires dilute quanta ($n_0 \ll 1/2$, so that the hard-core saturation $1 - 2n$ is negligible) and many of them per instability wavelength. Illustratively, $N_q = n_0 L_A \gtrsim 10$ at $n_0 = 0.1$ requires $L_A \approx 100$ sublattice sites, i.e. a chain of $N \approx 199$ qubits. That is two orders of magnitude beyond the lattices on which quantitative emulation is possible.

## 7 Computational cost and where quantum hardware would matter

**Classical emulation.** A full-statevector Trotter step of $H_0 + H_{ZZ}$ costs $0.112\,\text{ms}$ for $N = 11$ and $858\,\text{ms}$ for $N = 23$ on one core (Fig. 6b). This is a growth factor of $2.24$ per added qubit, i.e. $\mathcal{O}(N2^N)$ time and $16 \cdot 2^N$ bytes of memory (128 MiB at $N = 23$, 17.6 TB at $N = 40$). Sector-wise exact diagonalisation, used for our cross-checks, takes $0.2\,\text{s}$ at $N = 11$ and $11.4\,\text{s}$ at $N = 13$. The complete pre-registered sweeps took $3.8\,\text{s}$ and $8.0\,\text{s}$ of wall-clock time. The primary sweep comprises 25 (coupling, sideband) pairs, each run in four emulation variants plus the NLSE solution, on 30 cores. The secondary sweep comprises 6 couplings, 3 evolution variants and 4 pump densities. The classical PDE reference is cheaper still: an RK4 step of the discretised NLSE costs $38\,\mu\text{s}$ for 6 grid points and $3067\,\mu\text{s}$ for $6 \times 10^4$, i.e. linear in the grid size.

**Implication for quantum advantage.** The statement that $ZZ$ terms make the plasma-wave chain classically hard [14] remains true for generic states and observables; even linear oscillator dynamics admits exponential quantum speedups for suitably posed questions [61]. Our results sharpen what a quantum device would add for the physics studied here.

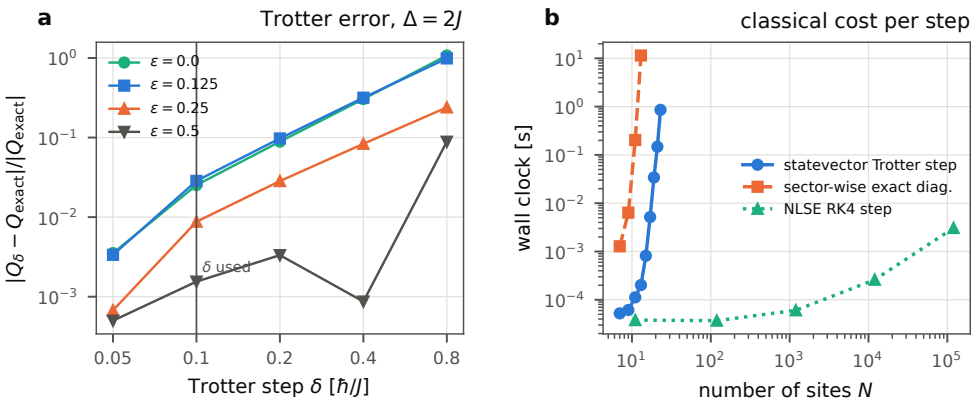



**Figure 6: a** Relative error of the emulated Kerr coefficient against Trotter step, relative to exact evolution. The vertical line marks the step used. **b** Classical wall-clock time per step on one core. Full-statevector Trotter step versus chain length $N$; sector-wise exact diagonalisation; and an RK4 step of the discretised NLSE on $(N+1)/2$ grid points.

- In the regime where the NLSE becomes quantitatively valid ($N \approx 199$ in the estimate of Sec. 6), statevector emulation is impossible: $1.3 \times 10^{61}$ bytes. However, the mean-field dynamics there is predicted by the NLSE itself at negligible cost.
- The quantum computer's distinctive contribution is the *corrections* to mean-field nonlinear plasma dynamics: few-quanta dilution, Kerr granularity and bound clusters. These dominate completely at the sizes accessible today.

On hardware, each Trotter step needs $N-1$ $XY$ gates and $N-2$ next-nearest-neighbour controlled phases. The latter are depth-two on a zig-zag ladder and need SWAP networks on a line. Resolving the virtual inter-branch processes requires $2\Delta\delta \ll 1$, and the MI windows of Sec. 4 span $11.8$–$87.4\,\hbar/J$, i.e. $10^2$–$10^3$ steps of $\delta = 0.1$. Such circuits are far deeper than the 12-step circuits run in Ref. [14].

## 8 Limitations and conclusions

**What the model captures.** The normal-ordered, density-dependent coupling of Eq. (4) turns the linear plasma-wave chain into a lattice realisation of the focusing NLSE, with explicitly derived coefficients (Table 1).

- It preserves the linear physics exactly at every $\epsilon$.
- It realises the nonlinear dispersion relation $\omega = \Delta + PK^2 + Q(\epsilon)n$ of a cutoff wave, and with it the self-phase modulation of an intense EM wave near $\omega_p$, quantitatively when $|V|t \lesssim 1$.
- This includes a defocusing kinematic contribution $J^2/2\Delta$ that the naive classical spin limit misses and that sets the threshold $\epsilon_c = J^2/4\Delta^2$ for the focusing (MI-prone) regime.

**What it does not capture at accessible sizes.** The hallmark consequence of that nonlinearity, modulational instability, does not appear in exact emulation of nine- to eleven-qubit chains. This is not because of noise, Trotter error or a mis-derived coefficient: the NLSE solved on the same grid grows, and the emulated Kerr coefficient matches. It is because a lattice holding $\sim 1$ wave quantum is far from the classical-field limit that MI presupposes. Most quanta have no partner, every number sector is a small Hermitian problem that cannot amplify, the pump dephases within an e-fold, and the per-pair $ZZ$ phase exceeds a radian before any amplification. At the most unstable lattice sidebands the local NLSE is moreover an inaccurate classical description of the lattice itself (Sec. 6).

**Limitations.**

- We studied a single committed coupling matrix, one plasma density ($\Delta = 2J$), one pump amplitude and product-state pumps. Number-state or squeezed pumps, other $\Delta$, and longer-range kernels were not emulated.

- The pre-registered comparison of the linear-limit dispersion with the reference paper's noiseless markers could not be met. We show that the markers are inconsistent with any ideal Trotter circuit at $\Delta = 0$, but we cannot reproduce the unpublished compiled-circuit details.
- The scaling argument for the onset of the NLSE regime rests on exact data for $N \leq 11$ and on analytic bounds, not on emulation of large chains. Truncated-Wigner or tensor-network studies [42, 43] of the crossover in $N_q$ are a natural next step.
- The mapping is derived to $\mathcal{O}(J^2/\Delta)$ and to leading order in $\epsilon n$. Hardware noise was not modelled.

**Outlook.** Two routes could bring NLSE-type instabilities within reach.

- *Longer-range kernels.* Coupling each site to $z \gg 1$ partners, the lattice analogue of a nonlocal Kerr response [26], reduces the granularity $|V|/\Gamma$ by $z$.
- *Larger lattices.* Lattices with $N \sim 10^2$ provide the required number of quanta.

Conversely, the few-quanta regime explored here, with its bound magnon clusters (quantum solitons) and collapse–revival dynamics, is where a quantum processor goes beyond the classical nonlinear PDE rather than merely reproducing it.

## References


[1] P. K. Shukla, N. N. Rao, M. Y. Yu, and N. L. Tsintsadze. Relativistic nonlinear effects in plasmas. *Phys. Rep.*, 138:1, 1986.

[2] P. Sprangle, E. Esarey, and A. Ting. Nonlinear theory of intense laser-plasma interactions. *Phys. Rev. Lett.*, 64:2011, 1990.

[3] Claire E. Max, Jonathan Arons, and A. Bruce Langdon. Self-modulation and self-focusing of electromagnetic waves in plasmas. *Phys. Rev. Lett.*, 33:209, 1974.

[4] S. E. Harris. Electromagnetically induced transparency in an ideal plasma. *Phys. Rev. Lett.*, 77: 5357, 1996.

[5] I. Y. Dodin and E. A. Startsev. On applications of quantum computing to plasma simulations. *Phys. Plasmas*, 28:092101, 2021.

[6] Ilon Joseph, Yuan Shi, Maxwell D. Porter, Alessandro R. Castelli, Vasily I. Geyko, Frank R. Graziani, Stephen B. Libby, and Jonathan L. DuBois. Quantum computing for fusion energy science applications. *Phys. Plasmas*, 30:010501, 2023.

[7] Alexander Engel, Graeme Smith, and Scott E. Parker. Quantum algorithm for the Vlasov equation. *Phys. Rev. A*, 100:062315, 2019.

[8] George Vahala, Linda Vahala, Min Soe, and Abhay K. Ram. Unitary quantum lattice simulations for Maxwell equations in vacuum and in dielectric media. *J. Plasma Phys.*, 86:905860518, 2020.

[9] Yuan Shi, Jianyuan Xiao, Hong Qin, and Nathaniel J. Fisch. Simulations of relativistic quantum plasmas using real-time lattice scalar QED. *Phys. Rev. E*, 97:053206, 2018.

[10] Stephen P. Jordan, Keith S. M. Lee, and John Preskill. Quantum algorithms for quantum field theories. *Science*, 336:1130, 2012.

[11] Christian W. Bauer, Zohreh Davoudi, Natalie Klco, and Martin J. Savage. Quantum simulation of fundamental particles and forces. *Nat. Rev. Phys.*, 5:420, 2023.

[12] Yuan Shi, Alessandro R. Castelli, Xian Wu, Ilon Joseph, Vasily Geyko, Frank R. Graziani, Stephen B. Libby, Jeffrey B. Parker, Yaniv J. Rosen, Luis A. Martinez, and Jonathan L. DuBois. Simulating non-native cubic interactions on noisy quantum machines. *Phys. Rev. A*, 103:062608, 2021.

[13] Yuan Shi, Bram Evert, Amy F. Brown, Vinay Tripathi, Eyob A. Sete, Vasily Geyko, Yujin Cho, Jonathan L. DuBois, Daniel Lidar, Ilon Joseph, and Matt Reagor. Simulating nonlinear optical processes on a superconducting quantum device. *J. Plasma Phys.*, 90:805900602, 2024.

[14] Bhuvanesh Sundar, Bram Evert, Vasily Geyko, Andrew Patterson, Ilon Joseph, and Yuan Shi. Simulating plasma wave propagation on a superconducting quantum chip. *Phys. Rev. Applied*, 25:024077, 2026. arXiv:2507.09479.

[15] M. J. Lighthill. Contributions to the theory of waves in non-linear dispersive systems. *IMA J. Appl. Math.*, 1:269, 1965.

[16] T. Brooke Benjamin and J. E. Feir. The disintegration of wave trains on deep water. *J. Fluid Mech.*, 27:417, 1967.

[17] V. E. Zakharov. Stability of periodic waves of finite amplitude on the surface of a deep fluid. *J. Appl. Mech. Tech. Phys.*, 9:190, 1968.

[18] V. E. Zakharov. Collapse of Langmuir waves. *Sov. Phys. JETP*, 35:908, 1972.

[19] V. I. Karpman and E. M. Krushkal. Modulated waves in nonlinear dispersive media. *Sov. Phys. JETP*, 28:277, 1969.

[20] Akira Hasegawa. Observation of self-trapping instability of a plasma cyclotron wave in a computer experiment. *Phys. Rev. Lett.*, 24:1165, 1970.

[21] A. I. Akhiezer and R. V. Polovin. Theory of wave motion of an electron plasma. *Sov. Phys. JETP*, 3:696, 1956.

[22] P. Kaw and J. Dawson. Relativistic nonlinear propagation of laser beams in cold overdense plasmas. *Phys. Fluids*, 13:472, 1970.

[23] V. E. Zakharov and L. A. Ostrovsky. Modulation instability: The beginning. *Physica D*, 238: 540, 2009.

[24] Catherine Sulem and Pierre-Louis Sulem. *The Nonlinear Schrödinger Equation: Self-Focusing and Wave Collapse*. Springer, New York, 1999.

[25] Yuri S. Kivshar and Michel Peyrard. Modulational instabilities in discrete lattices. *Phys. Rev. A*, 46:3198, 1992.

[26] Wiesław Krolikowski, Ole Bang, Jens Juul Rasmussen, and John Wyller. Modulational instability in nonlocal nonlinear Kerr media. *Phys. Rev. E*, 64:016612, 2001.

[27] T. Holstein and H. Primakoff. Field dependence of the intrinsic domain magnetization of a ferromagnet. *Phys. Rev.*, 58:1098, 1940.

[28] Takeo Matsubara and Hirotsugu Matsuda. A lattice model of liquid helium, I. *Prog. Theor. Phys.*, 16:569, 1956.

[29] P. Jordan and E. Wigner. Über das Paulische äquivalenzverbot. *Z. Phys.*, 47:631, 1928.

[30] A. Luther. Eigenvalue spectrum of interacting massive fermions in one dimension. *Phys. Rev. B*, 14:2153, 1976.

[31] Sidney Coleman. Quantum sine-Gordon equation as the massive Thirring model. *Phys. Rev. D*, 11:2088, 1975.

[32] Thierry Giamarchi. *Quantum Physics in One Dimension*. Oxford University Press, 2004.

[33] A. M. Kosevich, B. A. Ivanov, and A. S. Kovalev. Magnetic solitons. *Phys. Rep.*, 194:117, 1990.

[34] Freeman J. Dyson. General theory of spin-wave interactions. *Phys. Rev.*, 102:1217, 1956.

[35] Elliott H. Lieb and Werner Liniger. Exact analysis of an interacting Bose gas. I. The general solution and the ground state. *Phys. Rev.*, 130:1605, 1963.

[36] J. B. McGuire. Study of exactly soluble one-dimensional N-body problems. *J. Math. Phys.*, 5: 622, 1964.

[37] Yinchieh Lai and H. A. Haus. Quantum theory of solitons in optical fibers. I. Time-dependent Hartree approximation. *Phys. Rev. A*, 40:844, 1989.

[38] H. Bethe. Zur Theorie der Metalle. I. Eigenwerte und Eigenfunktionen der linearen Atomkette. *Z. Phys.*, 71:205, 1931.

[39] Michael Wortis. Bound states of two spin waves in the Heisenberg ferromagnet. *Phys. Rev.*, 132:85, 1963.

[40] Takeshi Fukuhara, Peter Schauß, Manuel Endres, Sebastian Hild, Marc Cheneau, Immanuel Bloch, and Christian Gross. Microscopic observation of magnon bound states and their dynamics. *Nature*, 502:76, 2013.

[41] Markus Greiner, Olaf Mandel, Theodor W. Hänsch, and Immanuel Bloch. Collapse and revival of the matter wave field of a Bose–Einstein condensate. *Nature*, 419:51, 2002.

[42] Anatoli Polkovnikov. Phase space representation of quantum dynamics. *Ann. Phys.*, 325:1790, 2010.

[43] J. Schachenmayer, A. Pikovski, and A. M. Rey. Many-body quantum spin dynamics with Monte Carlo trajectories on a discrete phase space. *Phys. Rev. X*, 5:011022, 2015.

[44] Paul Niklas Jepsen, Jesse Amato-Grill, Ivana Dimitrova, Wen Wei Ho, Eugene Demler, and Wolfgang Ketterle. Spin transport in a tunable Heisenberg model realized with ultracold atoms. *Nature*, 588:403, 2020.

[45] A. Trombettoni and A. Smerzi. Discrete solitons and breathers with dilute Bose–Einstein condensates. *Phys. Rev. Lett.*, 86:2353, 2001.

[46] L. D. Carr and J. Brand. Spontaneous soliton formation and modulational instability in Bose–Einstein condensates. *Phys. Rev. Lett.*, 92:040401, 2004.

[47] Jason H. V. Nguyen, De Luo, and Randall G. Hulet. Formation of matter-wave soliton trains by modulational instability. *Science*, 356:422, 2017.

[48] P. J. Everitt, M. A. Sooriyabandara, M. Guasoni, P. B. Wigley, C. H. Wei, G. D. McDonald, K. S. Hardman, P. Manju, J. D. Close, C. C. N. Kuhn, S. S. Szigeti, Y. S. Kivshar, and N. P. Robins. Observation of a modulational instability in Bose–Einstein condensates. *Phys. Rev. A*, 96:041601(R), 2017.

[49] Jin-Peng Liu, Herman Øie Kolden, Hari K. Krovi, Nuno F. Loureiro, Konstantina Trivisa, and Andrew M. Childs. Efficient quantum algorithm for dissipative nonlinear differential equations. *Proc. Natl. Acad. Sci. USA*, 118:e2026805118, 2021.

[50] Seth Lloyd, Giacomo De Palma, Can Gokler, Bobak Kiani, Zi-Wen Liu, Milad Marvian, Felix Tennie, and Tim Palmer. Quantum algorithm for nonlinear differential equations, 2020. arXiv:2011.06571.

[51] Ilon Joseph. Koopman–von Neumann approach to quantum simulation of nonlinear classical dynamics. *Phys. Rev. Research*, 2:043102, 2020.

[52] Kaiwen Weng, Zhaoyuan Meng, and Guohui Hu. Quantum computing of the nonlinear Schrödinger equation via measurement-induced potential reconstruction, 2026. arXiv:2601.19184.

[53] Chenjia Zhu, Zhaoyuan Meng, Yousheng Zhang, and Yue Yang. Quantum algorithm for the nonlinear Schrödinger equation via the Lax-pair scattering, 2026. arXiv:2606.29276.

[54] Efstratios Koukoutsis, Kyriakos Hizanidis, Lucas I. Inigo Gamiz, Oscar Amaro, Christos Tsironis, Abhay K. Ram, and George Vahala. Symplectic perspective to quantum computing for Hamiltonian systems, 2026. arXiv:2604.10794.

[55] B. Foxen et al. Demonstrating a continuous set of two-qubit gates for near-term quantum algorithms. *Phys. Rev. Lett.*, 125:120504, 2020.

[56] J. R. Schrieffer and P. A. Wolff. Relation between the Anderson and Kondo Hamiltonians. *Phys. Rev.*, 149:491, 1966.

[57] Sergey Bravyi, David P. DiVincenzo, and Daniel Loss. Schrieffer–Wolff transformation for quantum many-body systems. *Ann. Phys.*, 326:2793, 2011.

[58] Masuo Suzuki. General theory of fractal path integrals with applications to many-body theories and statistical physics. *J. Math. Phys.*, 32:400, 1991.

[59] Andrew M. Childs, Yuan Su, Minh C. Tran, Nathan Wiebe, and Shuchen Zhu. Theory of Trotter error with commutator scaling. *Phys. Rev. X*, 11:011020, 2021.

[60] Y. Castin and R. Dum. Low-temperature Bose–Einstein condensates in time-dependent traps: Beyond the U(1) symmetry-breaking approach. *Phys. Rev. A*, 57:3008, 1998.

[61] Ryan Babbush, Dominic W. Berry, Robin Kothari, Rolando D. Somma, and Nathan Wiebe. Exponential quantum speedup in simulating coupled classical oscillators. *Phys. Rev. X*, 13: 041041, 2023.

# A Derivation details

**Linear limit (items A1–A4).** On explicit $8\times8$ matrices for an $(A,\cdot,A)$ triple, $K\,n_1n_3-\frac{K}{4}(Z_1Z_3-Z_1-Z_3+1)=0$ identically. $\lim_{\epsilon\to0}K_{1,3}=0$, and $H_{ZZ}$ annihilates $|000\rangle,|001\rangle,|010\rangle,|100\rangle$. Hence $H_{ZZ}(\epsilon=0)=0$ as an operator, and for all $\epsilon$ $H_{ZZ}$ vanishes on the zero- and one-quantum sectors.

**Hard-core form (B1) and Bloch spectrum (C1–C4).** $-\frac{J}{4}(X_1Y_2-Y_1X_2)=\frac{iJ}{2}(a_1^\dagger a_2-a_2^\dagger a_1)$. The Bloch matrix of a uniform plasma has eigenvalues $\pm\sqrt{\Delta^2+J^2\sin^2k}$, with $\omega''(0)/2=J^2/2\Delta$.

**Schrieffer–Wolff projection (D1–D6).** With $\langle j|H|j{+}1\rangle=iJ/2$ and the $A$–$B$ gap $2\Delta$:

- the $A\to B\to A'$ amplitude is $\langle j{+}2|H|j{+}1\rangle\langle j{+}1|H|j\rangle/2\Delta=-J^2/8\Delta\equiv-t$;
- the return trip contributes $+t$ per $B$ neighbour;
- the effective dispersion $\Delta+2t-2t\cos2K$ agrees with $\sqrt{\Delta^2+J^2\sin^2K}$ to $\mathcal{O}(J^2/\Delta)$.

The interaction-dependent corrections to the energy denominators are of relative order $\epsilon n$ and contribute $\mathcal{O}(t\epsilon n^2)$ to the frequency of the uniform wave. They are beyond the cubic order of the NLSE.

**Linearisation (E4–E5) and its check.** Write $\delta\psi=u+iv$ (in phase and in quadrature with the pump). Linearising the hard-core mean field gives $\dot u=\varepsilon_K(1-2n)v$ and $\dot v=-[\varepsilon_K(1-2n)+2G_K]u$, hence Eq. (7). Here $\delta n=2\psi_0\,\mathrm{Re}\,\delta\psi/(1-2n)$ follows from $|\psi|^2=n(1-n)$, and the factor $(1-2n)$ on the dispersion is the hard-core reduction of hopping into occupied sites. Integrating the classical-spin equations of $H_{\mathrm{eff}}$ on a 48-site ring (`verify_linearisation.py`) reproduces Eq. (7) to within $4.9\times10^{-5}$. The small-amplitude, small-$K$ limit (F1–F4) matches the NLSE (8) with $P=4t$ and $Q=4t+2V$. The first omitted term is $\frac{112}{3}tK^4|\psi_0|^2(V+2t)$, a derivative nonlinearity and lattice correction.

**Classical-spin limit of the full chain (G1).** For the uniform $A$-sublattice wave, the $B$ spins remain exactly along $+z$, because the transverse fields from the two $A$ neighbours cancel. The classical frequency is therefore $\Delta+2Vn$, without $Q_{\mathrm{kin}}$. In the emulation $Q_{\mathrm{kin}}$ appears at the predicted value (Sec. 5.2), confirming that it is a quantum (virtual-fluctuation) effect.

## B Pre-registration record and deviations

**Timeline** (UTC; SHA-256 digests in `logs/method_plan_commits.log`).

1. 2026-10-05 03:32:17: Stage 0. Coupling matrix, phenomenon, PDE family, observable class, tolerance. At that time only $H_0$ had been emulated.
2. 2026-10-05 03:36:18: Stage 0b. Derived coefficients and the exact reference formula.
3. Pilot studies (`pilot_study.py`, `pilot_study2.py`; $36 + 6$ configurations; logs kept unchanged in `logs/pilot/`). These confirmed $Q_{\mathrm{kin}}$ and found no sideband growth for $N = 9, 11$, $\Delta = 1, 2, 3$, $n_0 = 0.05$–$0.3$.
4. 2026-10-05 03:47:06: Stage 1. Protocol and grids for the MI test (unchanged in substance), plus the secondary Kerr-coefficient test, with its window equal to the one used in the pilot.
5. Validation runs. After the validation runs the diagonal-energy routine of the emulator was rewritten to stream over sites, to permit the cost study at $N \leq 23$. For every coupling of the sweeps its output is identical to the original dense form (largest difference $0$; `regression_diag_energy.py`). All validation and analysis logs were then regenerated with the final code by `run_all.sh`; the pilots were not re-run.

**Deviations and additions.**

- The secondary observable was added after the pilots, before its validation runs.
- A dispersion variant with $\delta = 0.4$ was added because the reference phase plots suggest that step.
- The mechanism analyses of Sec. 6 are post hoc.

**Alternatives considered and rejected before Stage 0** (no emulation was run with them).

- Nearest-neighbour $ZZ$: no Kerr effect at the cutoff (Sec. 2).
- Uniform (unstaggered) next-nearest-neighbour $ZZ$: it makes the negative-frequency branch defocusing, which breaks conjugation symmetry.
- Bare $Z_j Z_{j+2}$ without the single-$Z$ counterterms: it shifts the linear dispersion by $-K$ per quantum, so the one-quantum physics would depend on $\epsilon$.
- Electromagnetically induced transparency and stimulated Raman scattering as targets: these require three-wave (number-changing) couplings that a number-conserving $ZZ$ term cannot provide.

## C Supplementary tables

## D Full-Hilbert-space evidence

Every emulation propagates all $2^N$ amplitudes ($2048$ for $N = 11$).

- *Norm and sectors.* Over all primary runs the norm is conserved to $1.5 \times 10^{-13}$, and the excitation-number populations, which are conserved by $H$, drift by at most $7.2 \times 10^{-14}$.
- *Weight outside the one-quantum subspace.* The pump states carry $11.5\,\%$ of their weight in sectors with two or more quanta.
- *Where the $ZZ$ term acts.* After evolving to $t = 40\,\hbar/J$ at $\epsilon = 0.25$, the sector-resolved overlap between $H$ and $H_0$ evolutions is $1.000$ in the one-quantum sector, and $0.70$ and $0.44$ in the two- and three-quantum sectors.
- *Truncation counterfactual.* Projecting the same initial state onto the zero- and one-quantum subspace makes the $H$ and $H_0$ evolutions identical (largest difference $0$), against $0.177$ in the full space. A one-quantum truncation would therefore remove the nonlinearity entirely.

(`full_hilbert_evidence.py`.)

**Table 4: Dispersion at $\epsilon = 0$ ($N = 9$, Ramsey protocol, 12 steps).** Mode labels follow the reference figure. Exact: eigenvalues of the one-quantum Hamiltonian. "ref.": digitised noiseless markers of Ref. [14].

| $\Delta$ | $ka/\pi$ | exact | exact evol. | $\delta = 0.8$ | $\delta = 0.4$ | ref. |
|---|---|---|---|---|---|---|
| 0 | $-0.4$ | $+0.951$ | $+0.951$ | $+0.954$ | $+0.959$ | $+1.083$ |
| 0 | $-0.3$ | $-0.809$ | $-0.809$ | $-0.800$ | $-0.816$ | $-0.724$ |
| 0 | $-0.2$ | $+0.588$ | $+0.588$ | $+0.595$ | $+0.619$ | $+0.788$ |
| 0 | $-0.1$ | $-0.309$ | $-0.309$ | $-0.340$ | $-0.304$ | $-0.201$ |
| 0 | 0.0 | 0.000 | 0.000 | 0.000 | 0.000 | $+0.089$ |
| 0 | $+0.1$ | $+0.309$ | $+0.309$ | $+0.340$ | $+0.304$ | $+0.358$ |
| 0 | $+0.2$ | $-0.588$ | $-0.588$ | $-0.595$ | $-0.619$ | $-0.417$ |
| 0 | $+0.3$ | $+0.809$ | $+0.809$ | $+0.800$ | $+0.816$ | $+0.924$ |
| 0 | $+0.4$ | $-0.951$ | $-0.951$ | $-0.954$ | $-0.959$ | $-0.875$ |
| $J/4$ | $-0.4$ | $+0.983$ | $+0.983$ | $+1.015$ | $+1.011$ | $+1.073$ |
| $J/4$ | $-0.3$ | $-0.847$ | $-0.847$ | $-0.798$ | $-0.827$ | $-0.706$ |
| $J/4$ | $-0.2$ | $+0.639$ | $+0.639$ | $+0.719$ | $+0.678$ | $+0.741$ |
| $J/4$ | $-0.1$ | $-0.397$ | $-0.397$ | $-0.327$ | $-0.420$ | $-0.297$ |
| $J/4$ | 0.0 | $+0.250$ | $+0.250$ | $+0.275$ | $+0.261$ | $+0.348$ |
| $J/4$ | $+0.1$ | $+0.397$ | $+0.397$ | $+0.454$ | $+0.402$ | $+0.481$ |
| $J/4$ | $+0.2$ | $-0.639$ | $-0.639$ | $-0.579$ | $-0.640$ | $-0.484$ |
| $J/4$ | $+0.3$ | $+0.847$ | $+0.847$ | $+0.885$ | $+0.877$ | $+0.942$ |
| $J/4$ | $+0.4$ | $-0.983$ | $-0.983$ | $-0.949$ | $-0.981$ | $-0.918$ |

**Table 5: All 25 (coupling, sideband) pairs of the MI test.** $\Gamma_{\mathrm{NLSE}}$: pre-registered discretised NLSE. $\Gamma_{\mathrm{E5}}$: exact-amplitude classical-spin limit with the lattice kernel. "NLSE num.": fit to the numerically solved NLSE. $g$: fitted $\cosh(\sqrt{g}\,t)$ parameter for the emulation ($\delta = 0.1$), exact evolution and $H_0$ ($g < 0$ means oscillation, not growth). $R(T)$: emulated response at the end of the window.

| $\epsilon$ | $p$ | $\Gamma_{\mathrm{NLSE}}$ | $\Gamma_{\mathrm{E5}}$ | NLSE num. | $g_{\mathrm{emu}}$ | $g_{\mathrm{exact}}$ | $g_{H_0}$ | $R(T)$ |
|---|---|---|---|---|---|---|---|---|
| 0.125 | 1 | 0.0236 | 0.0193 | 0.0236 | $-1.8 \times 10^{-4}$ | $-1.5 \times 10^{-4}$ | $-4.8 \times 10^{-4}$ | 0.42 |
| 0.125 | 2 | 0.0000 | 0.0000 | 0.0000 | $-2.4 \times 10^{-3}$ | $-2.5 \times 10^{-3}$ | $-4.3 \times 10^{-3}$ | $-0.21$ |
| 0.125 | 3 | 0.0000 | 0.0000 | 0.0000 | $-1.6 \times 10^{-2}$ | $-1.6 \times 10^{-2}$ | $-1.7 \times 10^{-2}$ | 0.17 |
| 0.125 | 4 | 0.0000 | 0.0000 | 0.0000 | $-3.5 \times 10^{-2}$ | $-3.4 \times 10^{-2}$ | $-3.3 \times 10^{-2}$ | 0.24 |
| 0.125 | 5 | 0.0000 | 0.0000 | 0.0000 | $-4.9 \times 10^{-2}$ | $-5.0 \times 10^{-2}$ | $-4.6 \times 10^{-2}$ | $-0.37$ |
| 0.1875 | 1 | 0.0373 | 0.0320 | 0.0373 | $-2.8 \times 10^{-4}$ | $-2.2 \times 10^{-4}$ | $-7.1 \times 10^{-4}$ | 0.65 |
| 0.1875 | 2 | 0.0484 | 0.0000 | 0.0484 | $-2.2 \times 10^{-3}$ | $-2.3 \times 10^{-3}$ | $-4.2 \times 10^{-3}$ | $-0.22$ |
| 0.1875 | 3 | 0.0000 | 0.0000 | 0.0000 | $-1.5 \times 10^{-2}$ | $-1.4 \times 10^{-2}$ | $-1.7 \times 10^{-2}$ | 0.06 |
| 0.1875 | 4 | 0.0000 | 0.0000 | 0.0000 | $-2.9 \times 10^{-2}$ | $-2.9 \times 10^{-2}$ | $-3.3 \times 10^{-2}$ | 0.43 |
| 0.1875 | 5 | 0.0000 | 0.0000 | 0.0000 | $-4.6 \times 10^{-2}$ | $-4.7 \times 10^{-2}$ | $-4.6 \times 10^{-2}$ | $-0.46$ |
| 0.25 | 1 | 0.0472 | 0.0410 | 0.0471 | $-3.5 \times 10^{-4}$ | $-3.0 \times 10^{-4}$ | $-7.8 \times 10^{-4}$ | 0.71 |
| 0.25 | 2 | 0.0740 | 0.0177 | 0.0739 | $-2.9 \times 10^{-3}$ | $-2.8 \times 10^{-3}$ | $-4.9 \times 10^{-3}$ | 0.15 |
| 0.25 | 3 | 0.0559 | 0.0000 | 0.0552 | $-1.6 \times 10^{-2}$ | $-1.5 \times 10^{-2}$ | $-1.7 \times 10^{-2}$ | $-0.03$ |
| 0.25 | 4 | 0.0000 | 0.0000 | 0.0000 | $-3.0 \times 10^{-2}$ | $-3.0 \times 10^{-2}$ | $-3.3 \times 10^{-2}$ | 0.41 |
| 0.25 | 5 | 0.0000 | 0.0000 | 0.0000 | $-4.6 \times 10^{-2}$ | $-4.7 \times 10^{-2}$ | $-4.6 \times 10^{-2}$ | $-0.37$ |
| 0.375 | 1 | 0.0625 | 0.0546 | 0.0623 | $-4.6 \times 10^{-4}$ | $-4.0 \times 10^{-4}$ | $-8.5 \times 10^{-4}$ | 0.78 |
| 0.375 | 2 | 0.1083 | 0.0559 | 0.1080 | $-3.4 \times 10^{-3}$ | $-3.3 \times 10^{-3}$ | $-5.4 \times 10^{-3}$ | 0.46 |
| 0.375 | 3 | 0.1250 | 0.0000 | 0.1242 | $-1.1 \times 10^{-2}$ | $-1.1 \times 10^{-2}$ | $-1.6 \times 10^{-2}$ | $-0.08$ |
| 0.375 | 4 | 0.1083 | 0.0000 | 0.1083 | $-2.2 \times 10^{-2}$ | $-2.2 \times 10^{-2}$ | $-2.9 \times 10^{-2}$ | $-0.62$ |
| 0.375 | 5 | 0.0625 | 0.0000 | 0.0625 | $-4.5 \times 10^{-2}$ | $-4.6 \times 10^{-2}$ | $-4.6 \times 10^{-2}$ | 0.44 |
| 0.5 | 1 | 0.0747 | 0.0655 | 0.0744 | $-5.5 \times 10^{-4}$ | $-4.8 \times 10^{-4}$ | $-8.9 \times 10^{-4}$ | 0.82 |
| 0.5 | 2 | 0.1340 | 0.0771 | 0.1336 | $-3.8 \times 10^{-3}$ | $-3.6 \times 10^{-3}$ | $-5.6 \times 10^{-3}$ | 0.61 |
| 0.5 | 3 | 0.1677 | 0.0000 | 0.1666 | $-1.2 \times 10^{-2}$ | $-1.2 \times 10^{-2}$ | $-1.6 \times 10^{-2}$ | 0.28 |
| 0.5 | 4 | 0.1746 | 0.0000 | 0.1739 | $-2.5 \times 10^{-2}$ | $-2.6 \times 10^{-2}$ | $-3.1 \times 10^{-2}$ | $-0.22$ |
| 0.5 | 5 | 0.1650 | 0.0000 | 0.1635 | $-3.8 \times 10^{-2}$ | $-3.8 \times 10^{-2}$ | $-4.3 \times 10^{-2}$ | $-0.62$ |

# E Reproducibility

All code is in `proposal/code`; `run_all.sh` regenerates every log, figure and table. Every number in this paper is set by a `\val{...}` macro generated by `make_tables.py` from the JSON logs, with the source field recorded beside each definition in `logs/results_macros.tex`.